

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY

LONERE – RAIGAD – 402 103

Summer Semester Examination – May 2019

Course: B. Pharm.

Semester: IV

Subject with Subject Code: Physical Pharmaceutics-II (BP403T)

Date: 31/05/2019

Marks: 75

Duration: 3hrs

Time: 10.30 am to 1.30 pm



- Instructions:**
- i) All questions are compulsory
 - ii) Figures to the right indicate full marks
 - iii) Draw the diagrams or flow charts wherever necessary

Q1. Solve the Following Multiple Choice Questions (MCQs) 20 M

1. The Polyphasic system where at least one dimension of the dispersed system is measured between 1 nm and 1 μ m is called as.....
 - A. Suspension
 - B. Emulsion
 - C. Micromeritics
 - D. Colloidal dispersion
2. Accelerated stability testing is done to
 - A. Predict Shelf life of the formulation
 - B. Predict dissociation constant
 - C. Predict diffusion constant
 - D. Determine activation energy
3. is the method of combining negatively and positively charged hydrophilic colloids
 - A. Coacervation
 - B. Peptization
 - C. Flocculation
 - D. Coagulation
4.the Gold numberthe protective ability of colloids.
 - A. Higher, Lower
 - B. Lower, Higher
 - C. Lower, Lower
 - D. Both A & B
5. Flocculated suspensions exhibit the flow of a type.....
 - A. Dilatants
 - B. Newtonian
 - C. Plastic
 - D. Pseudoplastic
6. Strong beam of light passed through colloidal particle solution becomes visible as light streak is called as.....
 - A. Scattering effect
 - B. Electric effect
 - C. Tyndall Effect
 - D. All of the above

7. Which of the following is the identification test for emulsion
- Dilution Test
 - Cobalt Chloride test
 - Conductivity Test
 - All of the Above
8. For an ideal suspension, the sedimentation volume should be.....
- Equal to one
 - less than one
 - Equal to one or more than one
 - zero
9. Microemulsions are in appearance/nature.
- Transparent
 - Bulky
 - Cohesive
 - Adhesive
10. In Coulter-counter, as the particles travel through the orifice, the event that occurs is.....
- Conductance between the electrodes increases
 - Electronic scanners produce photographs for volume measurement
 - Resistance between the electrodes increases
 - Sedimentation increases
11. When the angle of repose is between $25-30^\circ$ it will indicate that the powder material is having Flow.
- Excellent
 - Good
 - Passable
 - Poor
12. Helium Pycnometer is used to determine
- Particle Size
 - Surface area
 - True density
 - Sedimentation rate
13. One centipoise is equal to..... poise
- 0.1
 - 0.01
 - 0.001
 - 1
14. Brook-field viscometer is an example of type:
- Cone and plate
 - Extrusion
 - Rotating sphere
 - Rotating spindle

15. The Distance between two tangents on opposite sides of the particle parallel to some fixed direction is
- Projected Area diameter
 - Martin Diameter
 - Feret's diameter
 - Stoke's Diameter
16. The molecularity of the chemical reaction is equal to the number of that must collide simultaneously to give reaction product.
- Molecules
 - Electrons
 - Fractional value
 - All of these
17. The Half life ($t_{1/2}$) of substance in first order kinetics is expressed by.....
- $0.301/K_1$
 - $0.693/K_1$
 - $0.105/K_1$
 - $K_1/0.693$
18. The unit of specific reaction rate constant for second order reaction is.....
- Litre Moles⁻¹ Second⁻²
 - Litre Moles⁻¹ Second⁻¹
 - Second⁻¹
 - Moles Litre⁻¹ Second⁻¹
19. A drug suspension decomposes by zero-order kinetics with a rate constant of 2 mg mL⁻¹ month⁻¹. If the initial concentration is 100 mg mL⁻¹, what is the shelf life?
- 3 months
 - 4 months
 - 5 months
 - 6 months
20. In an elastic member Stress is directly proportional to the Strain within elastic limit is...
- Heckel Plot
 - Hookes Law
 - Poisson's ratio
 - Stress

Q.No.2 Attempt any TWO questions (Answer 2 out of 3)

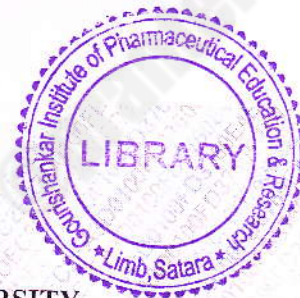
(20 Marks)

- What are Colloidal dispersion? Classify Colloids & compare their general properties. Discuss in detail Optical properties of Colloids.
- What is the concept of Rheology? Explain Newtonian and Non-Newtonian flow system with suitable examples.
- Define order of reaction. What are the methods to determine order of reaction? Explain in detail First order reaction rate constant derivation.

Q.No.3 Attempt any SEVEN questions (Answer 7 out of 9)**(35 Marks)**

1. Explain in brief DLVO theory for Stabilization of colloids.
2. Explain the concept of Donnan-membrane equilibrium.
3. Explain in detail about Thixotropy, measurement and its applications in pharmaceutical formulations.
4. Write a detail note on Derived properties of powders.
5. Explain in detail Accelerated stability studies & its significance in pharmacy.
6. Define and give importance of Micrometrics in pharmacy. Explain Sieving method in detail for determining particle size.
7. Explain Hydrolysis of drugs as degradation pathway with suitable examples. What are measures to be taken to prevent it?
8. Explain the concept of the use of controlled flocculation in case of flocculated suspension.
9. Differentiate between Plastic & Elastic deformation. Add a note on Heckel equation.

----- END OF PAPER -----



DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY

LONERE – RAIGAD – 402 103

Winter Semester Examination – December – 2019

Course: B.Pharm.

Semester: IV

Subject with Subject Code: Physical Pharmaceutics-II (BP403T)

Marks: 75

Date: 09/12/2019

Duration: 3hrs

Time: 10.30am to 1.30 pm

- Instructions: i) All questions are compulsory
ii) Figures to the right indicate full marks
iii) Draw the diagrams or flow charts wherever necessary.

Q.No.1 Attempt the following questions (All Questions Compulsory) (20 Marks)

- 1) Flocculated suspensions exhibit the flow of a type.....
 - A. Dilatants
 - B. Newtonian
 - C. Plastic
 - D. Pseudoplastic
- 2) Andreasen apparatus consists of.....
 - A. Balance
 - B. Electrode
 - C. Hydrometer
 - D. Pipette
- 3) The size of colloidal dispersed phase is
 - A. less than 1.0 nm
 - B. between 0.5 μm to 1.0 nm
 - C. greater than 0.5 μm
 - D. none of above
- 4) $K = x/at(a-x)$ is the rate constant associated with Order reaction.
 - A. Zero
 - B. First
 - C. Second
 - D. Pseudo first
- 5) A wetting agent is included in the formulation of a suspension, particularly when the suspended particles-----
 - A. Are hydrophobic
 - B. Are more denser than the vehicle
 - C. Are water soluble
 - D. Have lesser interfacial tension

- 6) is the irreversible type of colloids.
- Lyophilic colloids
 - Lyophobic colloids
 - Association colloids
 - Hydrophilic colloids
- 7) Which one of the following physical property is NOT a rheological property?
- Body and slip
 - Spreadability
 - Surface tension
 - Viscosity
- 8) When the angle of repose is between $25-30^\circ$ it will indicate that the powder material is having Flow.
- Excellent
 - Good
 - Passable
 - Poor
- 9) Porosity of a porous powder is defined as:
- void volume/Bulk volume
 - Void volume/bulk volume
 - Void volume/true volume
 - True volume/bulk volume
- 10) The degradation of solid in presence of moisture follows -----order kinetics.
- Zero
 - First
 - Second
 - Pseudo first
- 11) An 'emulsion within emulsion' is defined as-----
- $o/w/w$
 - $w/o/o$
 - $w/o/o/w$
 - $w/o/w$
- 12) require a specialized method for the preparation.
- Lyophilic colloids
 - Lyophobic colloids
 - Association colloids
 - Micelle

13) Brook-field viscometer is an example of type:

- A. Cone and plate
- B. Extrusion
- C. Rotating sphere
- D. Rotating spindle

14) In Coulter-counter, as the particles travel through the orifice, the event that occurs is.....

- A. Conductance between the electrodes increases
- B. Electronic scanners produce photographs for volume measurement
- C. Resistance between the electrodes increases
- D. Sedimentation increases

15) The HLB range of an emulsifier used in the preparation of W/O emulsion is--

- A. 3 to 6
- B. 7 to 12
- C. 13 to 15
- D. more than 15

16) The half life of substance in first order kinetics expressed by-----

- A. $0.301/K_1$
- B. $0.693/K_1$
- C. $0.105/K_1$
- D. $K_1/0.693$

17) The type of viscosity specified in I.P.(Ostwald viscometer) is:

- A. Absolute viscosity
- B. Dynamic viscosity
- C. Kinematic viscosity
- D. Viscosity coefficient

18) For an ideal suspension, the sedimentation volume should be-----

- A. Equal to one
- B. less than one
- C. Equal to one or more than one
- D. zero

19) In case of suspensions, all statements below are true EXCEPT that-----

- A. It contains a suspending agent
- B. It does not include a preservative
- C. It is an injectable preparation
- D. It is an oral preparation

20) -----shows fastest rate of degradation among dosage form.

- A. Solid
- B. Emulsion
- C. Suspension
- D. Solution

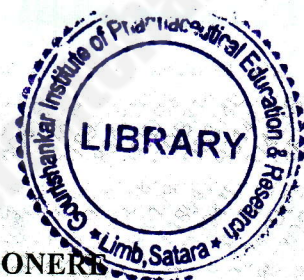
Q.No.2 Attempt any TWO questions of the following: (20 Marks)

- A) Explain Newtonian & Non Newtonian types of flow with its rheograms, mechanisms & with suitable examples.
- B) What are colloidal dispersion. Explain its types & Discuss electrical, kinetic & optical properties of colloids.
- C) Define emulsion, classify it with examples & Discuss in detail instabilities of emulsions & how is it prevented.

Q.No.3 Attempt any SEVEN questions of the following: (35 Marks)

- A) Define Micromeritics with its applications & discuss methods for determining particle size.
- B) Explain the influence of physical & chemical factors on degradation of Pharmaceuticals.
- C) Write the principle & working of Ostwald's viscometer & write applications of study of rheology.
- D) What is meant by sedimentation parameters? How are they evaluated?
- E) Define angle of repose? How it is determined? Give its importance in preparation of dosage forms.
- F) Write significance of accelerated stability testing & Discuss procedure for accelerated stability testing with any one example.
- G) Explain in detail about Thixotropy & Negative thixotropy with its curves.
- H) What is protective action of colloids. How it is determined & Enumerate the pharmaceutical applications of colloids.
- I) Define porosity with its applications in pharmacy & write note on different properties of powders.

*****END OF PAPER*****



DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERGAON
End Semester Examination – Summer 2022

Course	: B. Pharmacy (Second Year)	Sem.	: IV
Subject Name	: Physical Pharmaceutics-II	Subject Code	: BP403T
Max Marks	: 75	Date:30/08/2022	Duration : 3.45 Hr

Instructions –

1. All questions are compulsory
2. Answers to MCQs should be written in full sentences
3. Draw diagrams / figures wherever necessary
4. Figures to right indicate full marks

Q.No.1. Attempt the following Multiple choice questions. (MCQs) (20X 1= 20 Marks)
(All the questions are compulsory)

- Gold number is defined as number of milligram of protective colloid required in 10 mL of red gold sol to prevent the change in colour from red to violet on addition of 1 mL of..... Solution.**
 - a. 10 % NaCl
 - b. 0.1 % NaCl
 - c. 10 % KCl
 - d. 0.1 % KCl
- Sedimentation velocity of spherical particles is governed by.....**
 - a. Charle's Law
 - b. Stoke's Law
 - c. Osmotic Pressure
 - d. None of the above
- is the difference in the potential between the shear plane & the electroneutral region in the dispersion**
 - a. Peptization
 - b. Nernst potential
 - c. Electrokinetic Potential
 - d. Steady State
- For an ideal suspension, the sedimentation volume should be...**
 - a. Zero
 - b. Less than 1
 - c. More than 1
 - d. Equal to 1
-is the process of formation of light, fluffy aggregates held together by physical forces.**
 - a. Flocculation
 - b. Deflocculation
 - c. Coalescence
 - d. All of these
- η_{sp}/c is used for viscosity determination by using.....viscometer**
 - a. Capillary
 - b. Falling Sphere
 - c. Cup & Bob
 - d. Cone & Plate



a. Plastic b. Pseudoplastic
c. Dilatant d. Spurs

$$\mu = \frac{\text{Lateral Strain}}{\text{Linear Strain}}$$

- Elastic modulus
- Hookes law
- Heckel equation
- Poisson's ratio

The reciprocal of viscosity is known as.....

- Fluidity
- Mobility
- Ductility
- Plug flow

The change in velocity between two planes of liquid which is separated by distance is

a. Shearing stress b. Rate of shear

c. Strain d. Yield value

The unit of specific reaction rate constant for Second order reaction is

a. Litre Moles⁻¹ Second⁻² b. Litre Moles⁻¹ Second⁻¹
c. Second⁻¹ d. Moles Litre⁻¹ Second⁻¹

Wet gum method is also known as

a. Continental Method b. Forbe's method
c. English method d. Bottle method

A drug suspension decomposes by zero-order kinetics with a rate constant of 2 mg mL⁻¹ month⁻¹. If the initial concentration is 100 mg mL⁻¹, what is the shelf life?

a. 2 months b. 3 months
c. 4 months d. 5 months

Stability is the ability of a pharmaceutical product to retain the physical, chemical, microbiological and biopharmaceutical properties

a. Drug stability b. Chemical kinetics
c. Order of reaction d. Micromeritics

Hausner ratio is

- Tapped Density/Bulk density
- Bulk density/ Tapped Density
- Bulk Density/ Void volume
- Void volume/ Bulk Density

Which of the following is the identification test for emulsion

- Dilution Test
- Cobalt Chloride test
- Conductivity Test
- All of the Above

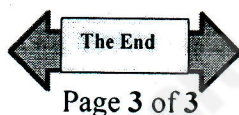
- xvii. The density of the dispersed phase is more than that of the dispersion medium. According to Stoke's equation, the creaming is:
- At the center of emulsion
 - In both the directions
 - In upward direction
 - In downward direction
- xviii. The rate equation for a chemical reaction is reported as $(-dc/dt)=kc$. The order of reaction is....
- Zero order
 - Pseudo Zero order
 - Pseudo first order
 - First order
- xix. The Carr's compressibility index value 26 - 31 indicate that the flow will be...
- Excellent
 - Passable
 - Good
 - Poor
- xx. In coulter-counter method, as the particles travelled through the orifice the event that occurs is...
- Conductance between the electrodes increases
 - Electronic scanners produces photographs for volume measurement
 - Resistance between the electrodes increase
 - Sedimentation increase

Q.No.2 Solve any TWO from following questions (Long Answers) (2 X 10 = 20 Marks)

- What is rheology and elaborate different types of flows in liquids.
- What are the properties of colloids? Explain electrical properties in detail.
- What is Molecularity & Order of Reaction? Explain the different methods used for determination of 'order of reaction'.

Q.No.3 Solve any SEVEN from following questions (Short Answers) (7 X 5 = 35 Marks)

- Define following terms:
a. Angle of repose b. Porosity, c. True density d. Electrophoresis e. Stability testing
- Explain the Van't Hoff equation.
- Describe different types of Particle diameters.
- Differentiate between Plastic & Elastic deformation. Add a note on Heckel equation.
- Elaborate the concept of Thixotropy.
- Derive the derivation of Zero order reaction rate constant.
- Explain the importance of Micrometrics in pharmacy with the help of suitable examples.
- Explain the concept of Controlled Flocculation.
- Write a detail note on Accelerated stability testing for pharmaceutical products.

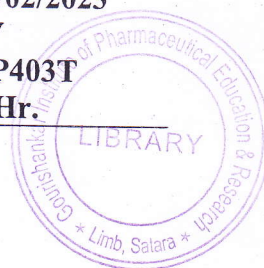


DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

End Semester Examination – Supplementary Winter 2022

Course : B. Pharmacy
Subject Name : Physical Pharmaceutics II
Max Marks : 75

Date : 11/02/2023
Sem : IV
Subject Code : BP403T
Duration : 3 Hr.



Instructions:

1. All questions are compulsory
2. Draw diagrams / figures wherever necessary
3. Figures to right indicate full marks

Q. 1. Objective Type Questions (Answer all the questions) (10 x 2) = 20

- i) Define angle of repose and give its importance.
- ii) Elaborate the concept of thixotropy.
- iii) Explain any 2 general characteristics of colloidal particles.
- iv) Distinguish between flocculated and deflocculated suspension.
- v) What is first order of reaction?
- vi) Draw HLB scale.
- vii) What measures you will take to improve the flow property of a granules?
- viii) Define gold no and Schulze hardy rule.
- ix) What is the principle of Coulter Counter?
- x) Explain the mechanism of micelles formation.

Q. 2. Long Answers (Answer 2 out of 3) (2 x 10) = 20

- i) What is drug stability? Explain in details accelerated stability testing.
- ii) Define and classify colloidal dispersion. Add a note on comparative properties of different colloidal system.
- iii) Elaborate any 4 derived properties of powder.

Q. 3. Short Answers (Answer 7 out of 9) (7 x 5) = 35

- i) Explain different factors influencing the chemical degradation of pharmaceutical product.
- ii) Differentiate between plastic and elastic deformation. Add a note on Heckel equation.
- iii) Write a note on Non-Newtonian systems
- iv) Enlist different methods to determine particle size and explain any one.
- v) Discuss sedimentation parameter.
- vi) Describe the electrical property of colloidal system.
- vii) Explain application of micrometrics in pharmacy.
- viii) Classify viscometer and explain working principle of Ostwald Viscometer.
- ix) What is emulsion? Explain stability of emulsion.

-----END OF THE PAPER-----

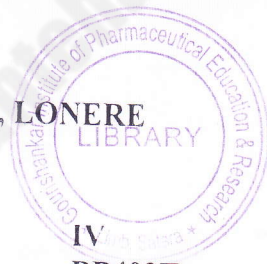
DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

End Semester Examination – Summer 23

Date: 14/07/2023

Course: B. Pharmacy
Subject Name: Physical Pharmaceutics-II
Max Marks: 75

Sem: IV
Subject Code: BP403T
Duration: 3 Hr.



Instructions:

1. All questions are compulsory
2. Draw diagrams / figures wherever necessary
3. Figures to right indicate full marks

Q. 1. Objective Type Questions (Answer all the questions)

(10 x 2) = 20

- i) What is Brownian movement? Which formulation shows Brownian movement?
- ii) Explain the term 'Colloid' mention its applications.
- iii) Explain the term 'Shear thinning' and 'Shear thickening' systems. Give one example of each.
- iv) Discuss the term 'Thixotropy'.
- v) Differentiate between flocculated and deflocculated suspension.
- vi) Describe different types of strain.
- vii) Enlist various identification tests for emulsion and explain any one of them.
- viii) Write steps involved in the preparation of emulsion by wet gum method.
- ix) Draw neat labelled diagram of Anderson pipette.
- x) Explain order of reaction with example.

Q. 2. Long Answers (Answer 2 out of 3)

(2 x 10) = 20

- i) Define Rheology and explain in details all fluids (flow) under rheology.
- ii) What is Micromeritics? Give its importance in pharmacy. Explain in details methods used to determine particle size.
- iii) What is colloidal dispersion? How will you stabilise colloidal dispersion by DLVO theory.

Q. 3. Short Answers (Answer 7 out of 9)

(7 x 5) = 35

- i) Write a note on chemical kinetics.
- ii) Explain in details about ICH guidelines for accelerated stability testing.
- iii) Define drug stability. Enlist and explain types of degradation of drug with example.
- iv) Define stress and differentiate between elastic and plastic deformation.
- v) Define viscosity. Classify viscometers and explain in details working of any one class of viscometers.
- vi) Explain general properties of different types of colloids.
- vii) Write a note on emulsion.
- viii) Discuss about coulter-counter method for determination of particle volume.
- ix) State and explain Hooke's Law.

-----END OF THE PAPER-----

DR BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

End Semester Supplementary Examination- Winter-2023

Date : 06/01/2024

Course : B.Pharmacy

Subject Name: Physical Pharmaceutics II

Max Marks : 75

Semester: IV

Subject Code: BP403T

Duration: 3Hr.

Instructions:

- 1. All questions are compulsory**
- 2. Draw diagrams/figures wherever necessary**
- 3. Figures to right indicate full marks**

1. Objective Type Questions (Answer all the questions) (10 x 2) = 20 Marks

- i. Define peptization with example.
- ii. State Bancroft's rule. Draw HLB scale.
- iii. Differentiate molecular dispersion and colloidal dispersion with example.
- iv. What do you mean by "Pseudo-Zero Order Kinetics"?
- v. Draw rheograms of thixotropic material showing bulge and spur.
- vi. Define angle of repose and give its importance.
- vii. Define porosity. Give applications in Pharmacy.
- viii. Distinguish between flocculated and deflocculated suspension.
- ix. Explain the need for stability testing?
- x. Define Viscosity and Fluidity with examples.

2. Long Answers (Answer 2 out of 3) (2 x 10) = 20 Marks

- i. Define first order reaction. Deduce an equation for determining the rate constant, half life and shelf-life of a zero order reaction.
- ii. Explain in detail about plastic flow, pseudo-plastic flow and dilatant flow with suitable graph.
- iii. Describe types of physical instability in emulsion. Differentiate between wet and dry gum methods for preparation of emulsion.

3. Short Answers (Answer 7 out of 9) (7 x 5) = 35 Marks

- i. Discuss different tests to evaluate the flowability of powder
- ii. Write in detail about the formulation of suspension.
- iii. Differentiate between lyophilic and lyophobic Colloids.
- iv. Explain sedimentation method of particle size determination alongwith its advantages and disadvantages.
- v. Explain the preventive measures for chemical degradation due to hydrolysis and oxidation.
- vi. Define :1.Surface diameter,2.Volume diameter,3.Projected diameter 4.Stoke's diameter & 5. Sieve diameter
- vii. Explain determination of viscosity by using multi point viscometer.
- viii. Describe Optical and Kinetics properties of colloids.
- ix. State Hooke's law.Differentiate between strain and stress.

-----END OF THE PAPER-----

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Regular/Supplementary Summer Examination – 2024

Course: B. Pharmacy

Semester: IV

Subject Name: Physical Pharmaceutics II

Subject Code: BP403T

Max Marks: 75

Date: 19-06-2024

Duration: 3 Hr.

Instructions to the Students:

1. All questions are compulsory
2. Draw diagrams / figures wherever necessary
3. Figures to right indicate full marks

Q.1. Objective Type Questions (Answer All the Questions) (10 X 2) = 20

- i) What is the molecularity of a reaction?
- ii) What is a gold number? Write down their significance.
- iii) Define following terms; a. stokes diameter b. projected diameter.
- iv) Explain the concept of bulges.
- v) What is dilatancy? Give the reasons for the dilatancy.
- vi) What is the order of reaction? Enlist the factors that affect the order of reaction.
- vii) What is the Hardy Schulze rule for colloids?
- viii) Define suspensions and write down the ideal properties of suspensions.
- ix) Define micromeritics write about the importance of micromeritics in pharmacy.
- x) Why do suspensions follow the zero order and tablets follow the order of reaction?

Q.2. Long Answers (Answer 2 out of 3) (10 X 2) = 20

- i) Describe the colloidal dispersion. Explain in detail the different properties of colloids
- ii) Define the emulsion and classify it. Discuss in detail the instability of emulsions and theories of emulsions.
- iii) Describe the different methods for determining the particle size using different methods. Discuss the advantages and disadvantages of each method for particle size determination.

Q.3. Short Answers (Answer 7 out of 9) (5 X 7) = 35

- i) Derive the equation for the first order of reaction.
- ii) Explain the different preparation and purification methods of colloids.
- iii) Differentiate between elastic deformation and plastic deformation. Explain the heckle equation.
- iv) List the factors that affect the viscosity of liquids. Explain a cup- and bob type viscometer.
- v) Illustrate the distinguishing features of flocculated and deflocculated suspensions.
- vi) Write notes on thixotropy and the measurement of thixotropy.
- vii) Describe the flow behaviour of shear thickening and shear thinning systems.
- viii) What is drug stability? Explain in detail about accelerated stability testing.
- ix) Write a note about the about the spreading coefficient.

*** END OF THE PAPER ***

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